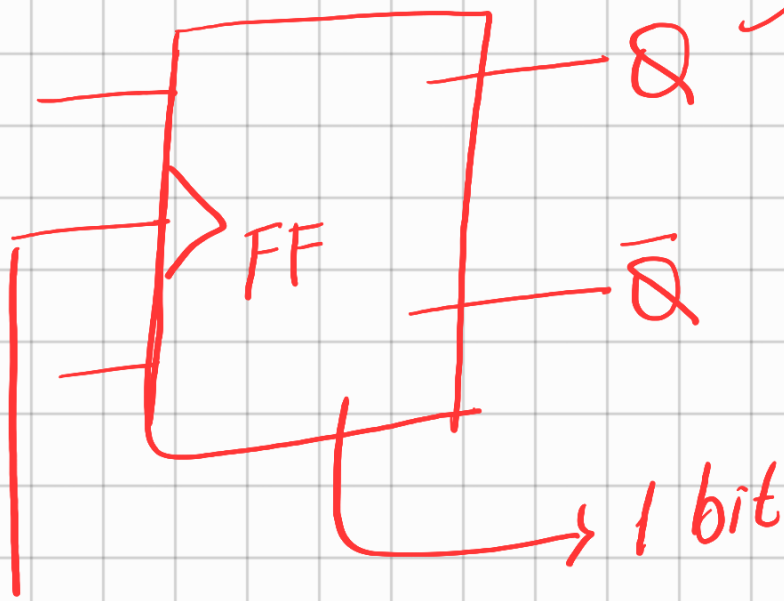
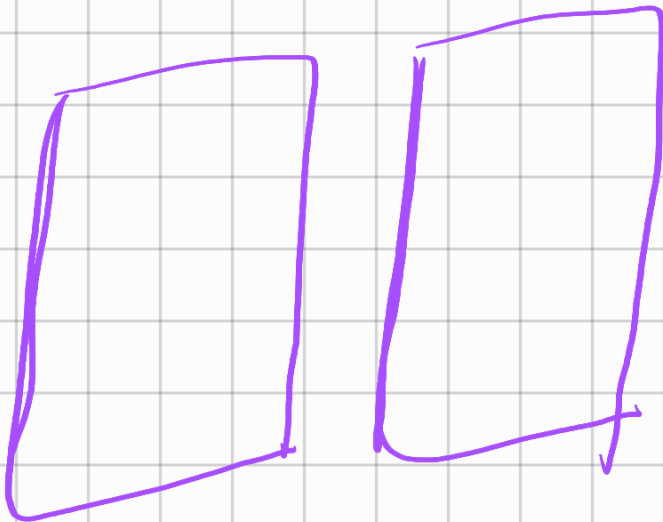


Flip-Flop



1 girişli
ve 2 girişli
olabilir

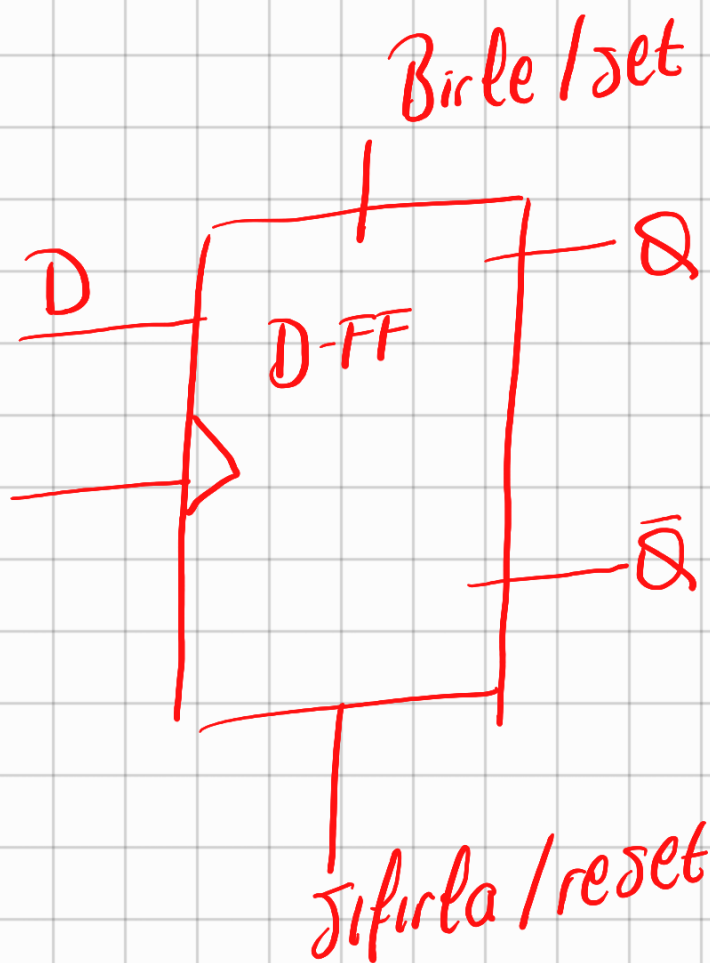


$2^3 \text{ bit} \rightarrow 8 \text{ farklı durum}$

*

DFF
JKFF
JRFF
TFF

DFF
Σ



$$Q(t+1) = D(t)$$

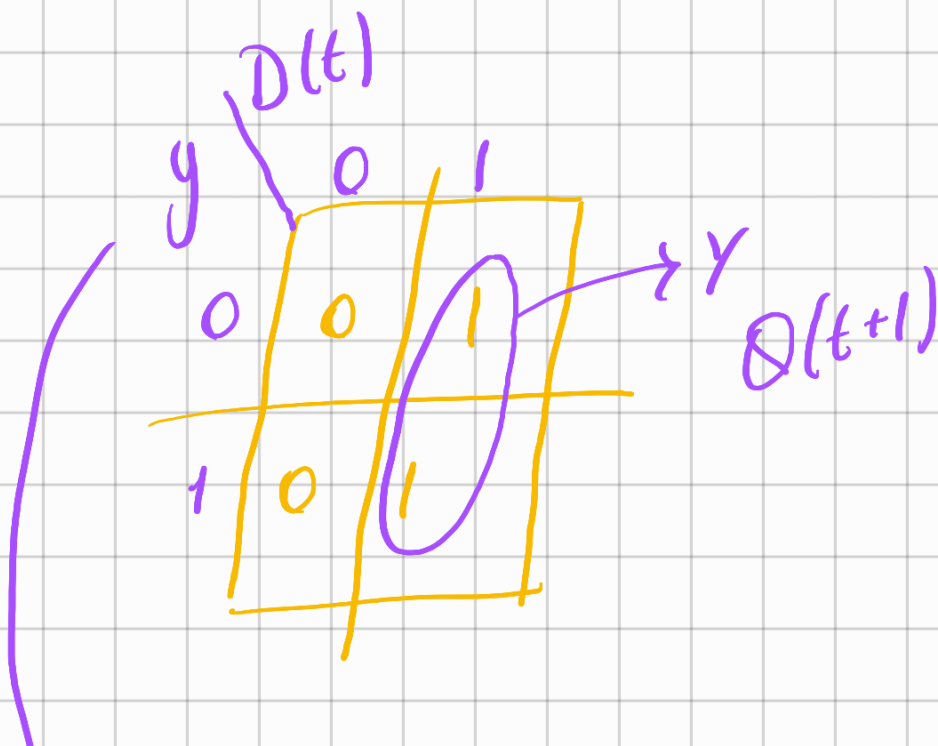




$$Q(t+1) = Y$$

$$Q(t) = y$$

giriş	saat	çıkışlar
D		Q $\bar{Q}$
0		0 1
1		1 0



$Q(t)$

Bunları
Bil

$$y=D$$

$Q(t) \quad Q(t+1)$

Uyarma
tablosu

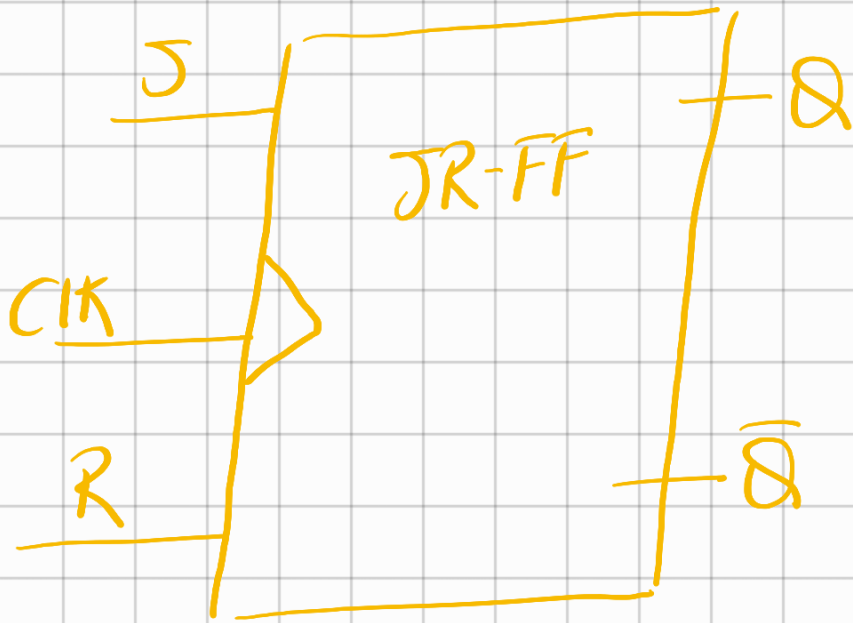
y'	00	01	11	10
D	0	1	1	0

JRFF

(set/reset
FF)



$$y = y_K + U$$



2 girişli
bir
flip-flop'tur

$$Q(t+1) = Q(t) \overline{R(t)} + J(t)$$

$R(t) J(t) = 0$ için tanımlı

$R(t) J(t) = 1$ için tanımsız

giriş JK	saat	çıkış Q $\bar{Q}$
00		Q $\bar{Q}$
00		01
10		10
11		tanımsız

JK

y	00	01	11	10
0	0	0	0	1
1	1	0	0	1

→ y

$$y = J + R'y$$

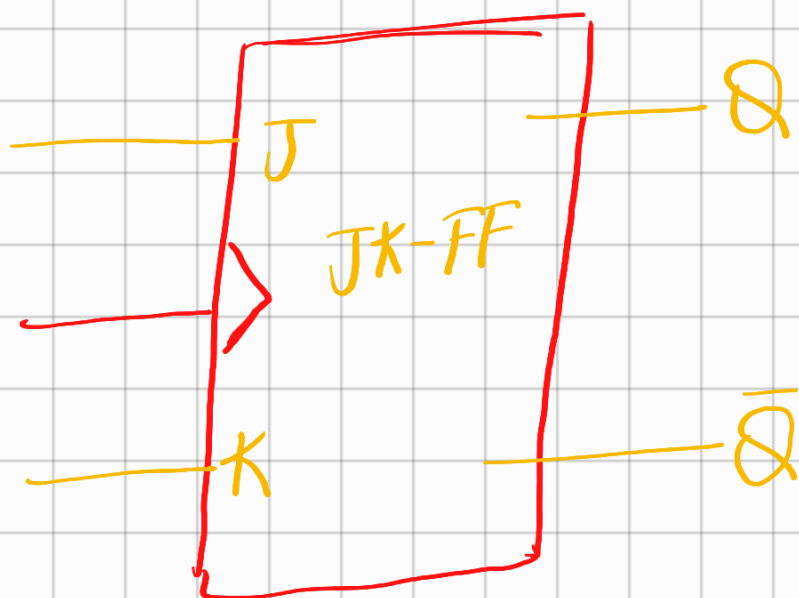
y	00	01	11	10
J	0	1	0	0
K	0	0	0	1

→ uyarma tablosu

$$y = Q(t)$$

$$y = Q(t+1)$$

JKFF



$$Q(t+1) = Q(t) \cdot K(t) + \bar{Q}(t) \cdot J(t)$$

$$\hookrightarrow yK' + Jy'$$

input J K	R	J00t		Q(t+1)	
		Q		Q̄	
00		□		□	
01		□		01	
10		□		10	
11		□		□	

→ 11

	JK	00	01	11	10
0		0	0	1	1
1		1	0	0	1

→ Y

$$Y = Jy' + Ky$$

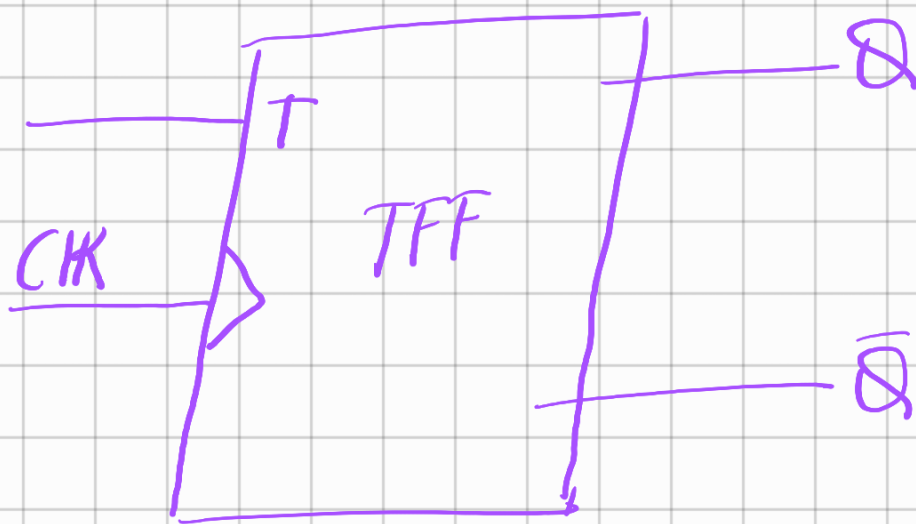
$t \rightarrow t+1$

$t \rightarrow t+1$

$t \rightarrow t+1$

Y	00	01	11	10
JK	00	10	00	01

T-FF (toggle)



$$Q(t+1) \rightarrow Q(t) \oplus T(+)$$

$$Y = y \oplus T$$

giriş T	saat 	çıkış $Q \bar{Q}$
0		$Q \bar{Q}$
1		$\bar{Q} Q$

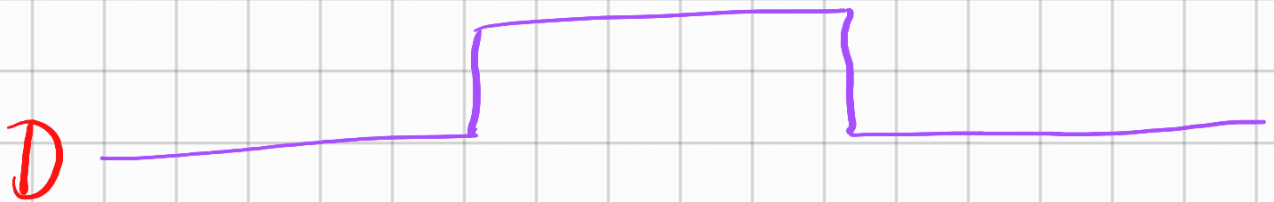


0	0	1
1	0	

$$Y = Ty' + T'y$$

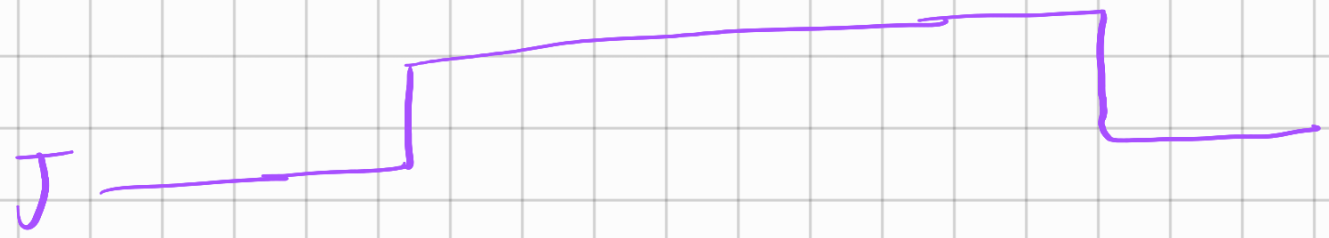
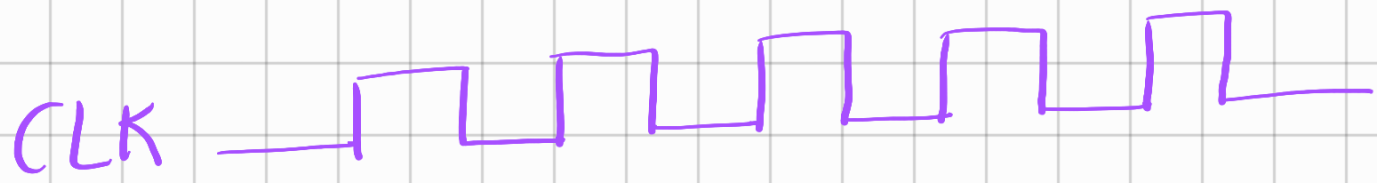
$$Y = T \oplus y$$

$$Y = \overline{D}$$



D-FF

JK-FF



K

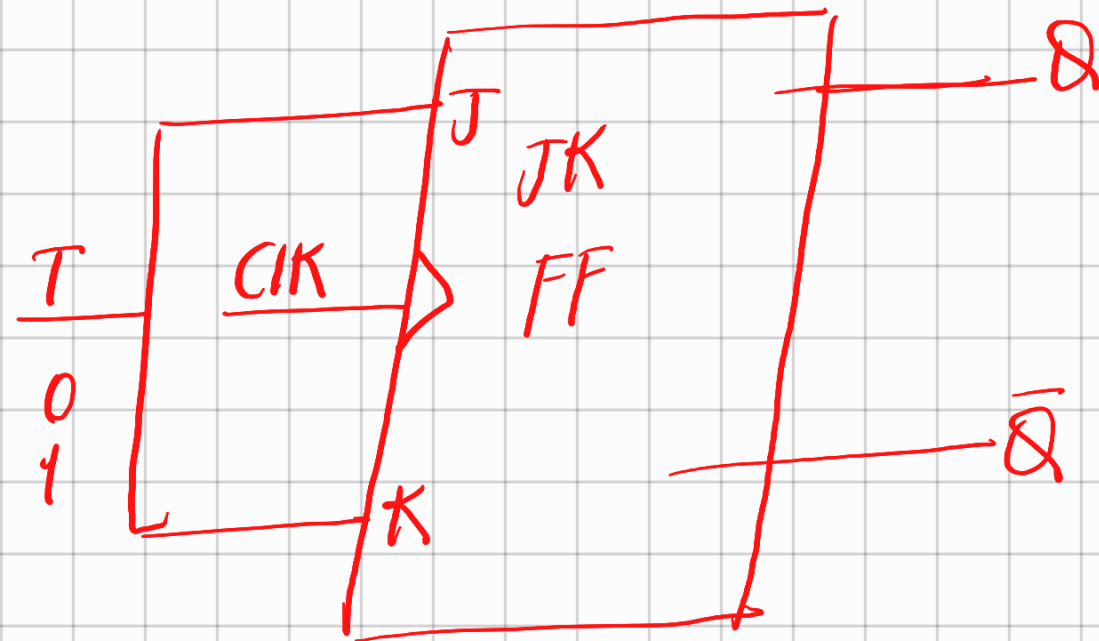
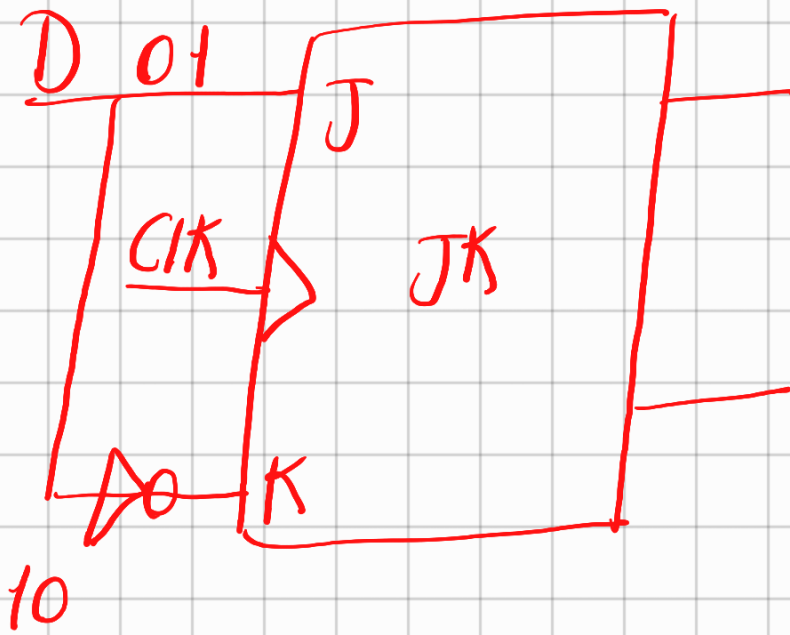
Q

$\bar{Q}$

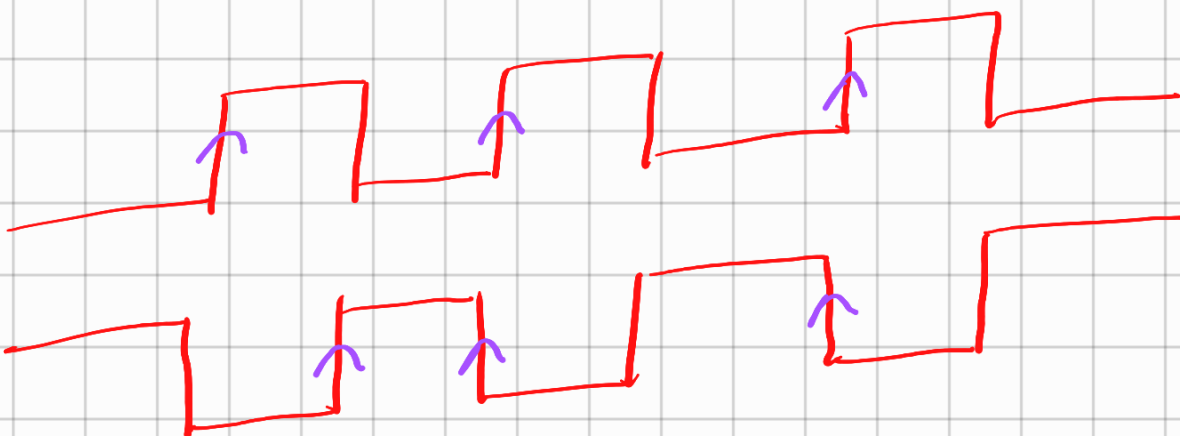
$$y = Jy' + K'y$$

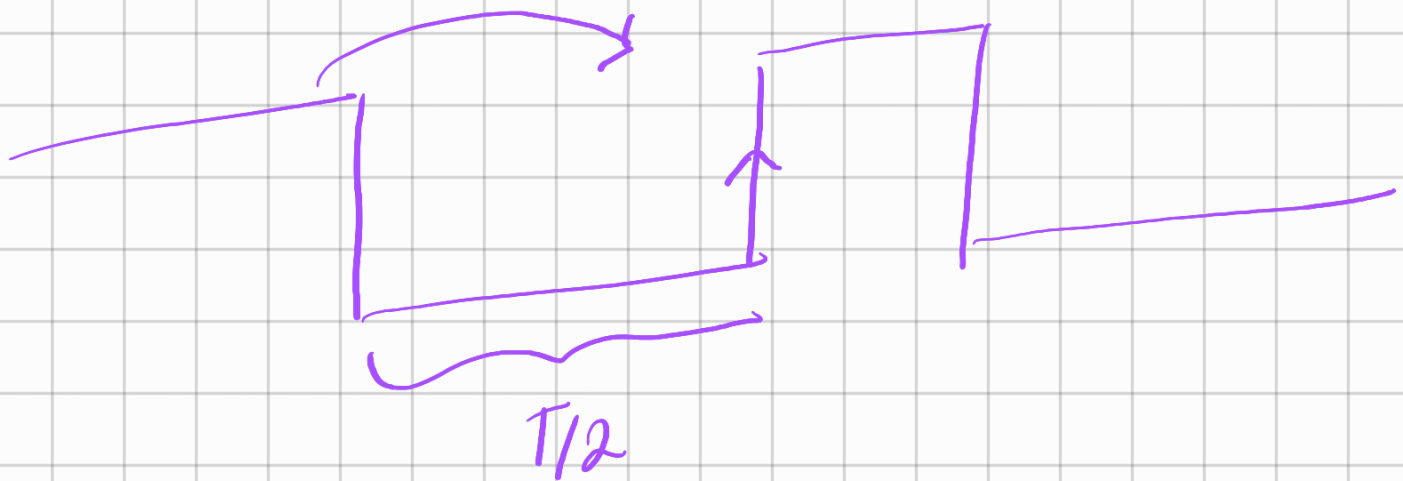
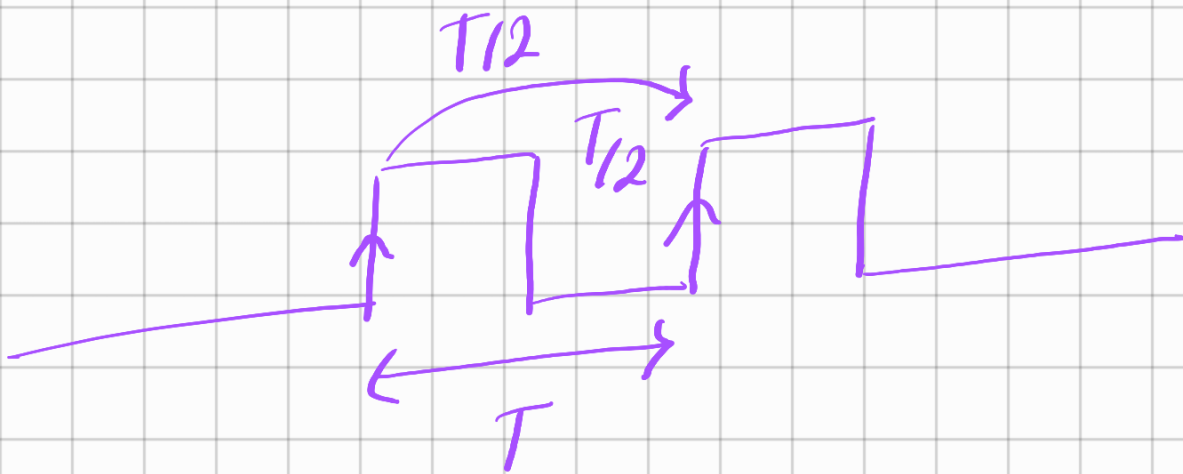
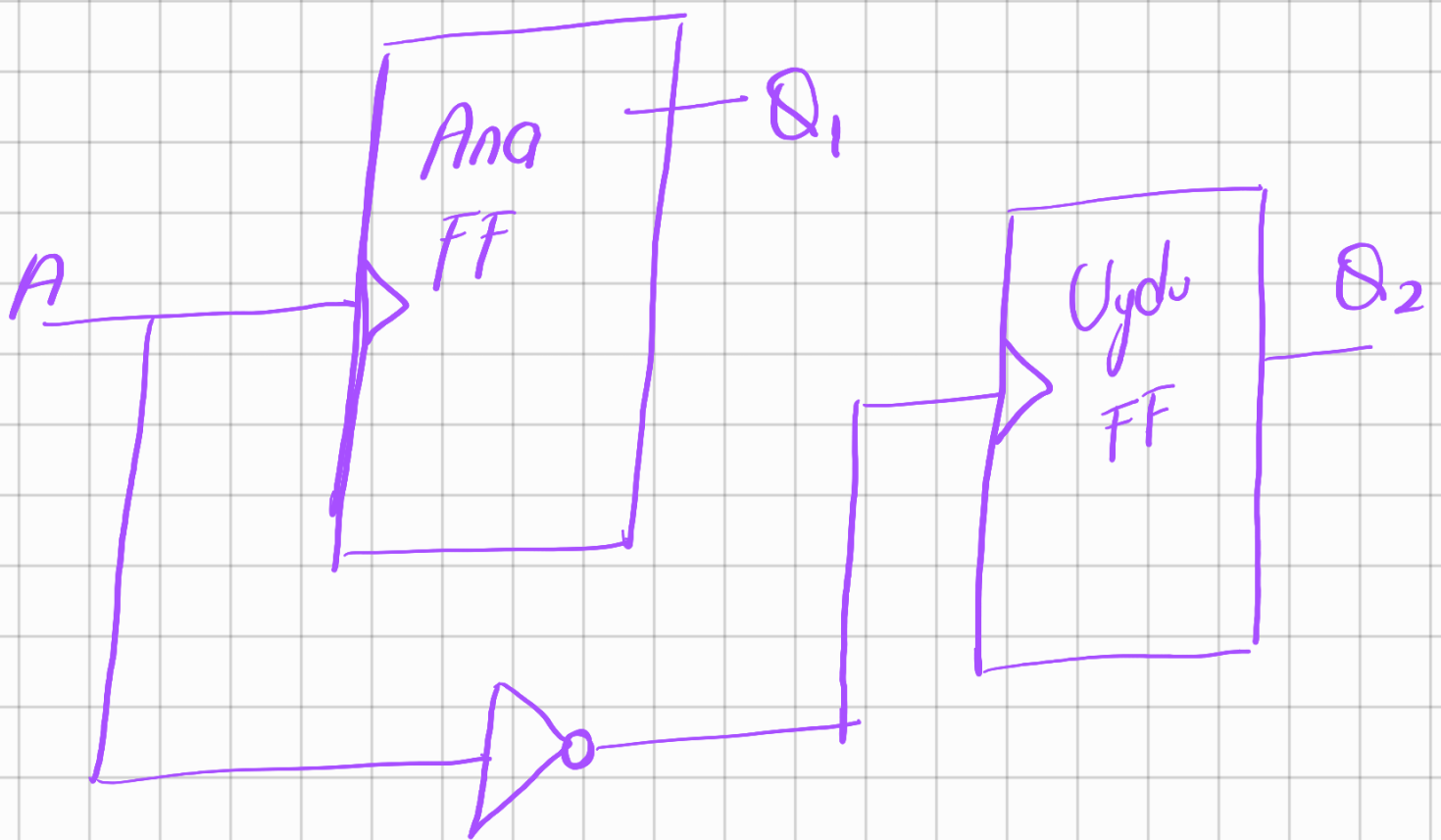
giriş JK	çıkış Q
00	Q
01	0
10	1
11	$\bar{Q}$

Örnek $\rightarrow$ JK-FF T ve D tipi FF yap-
mak mümkün müdür?



Ana Uydu FF

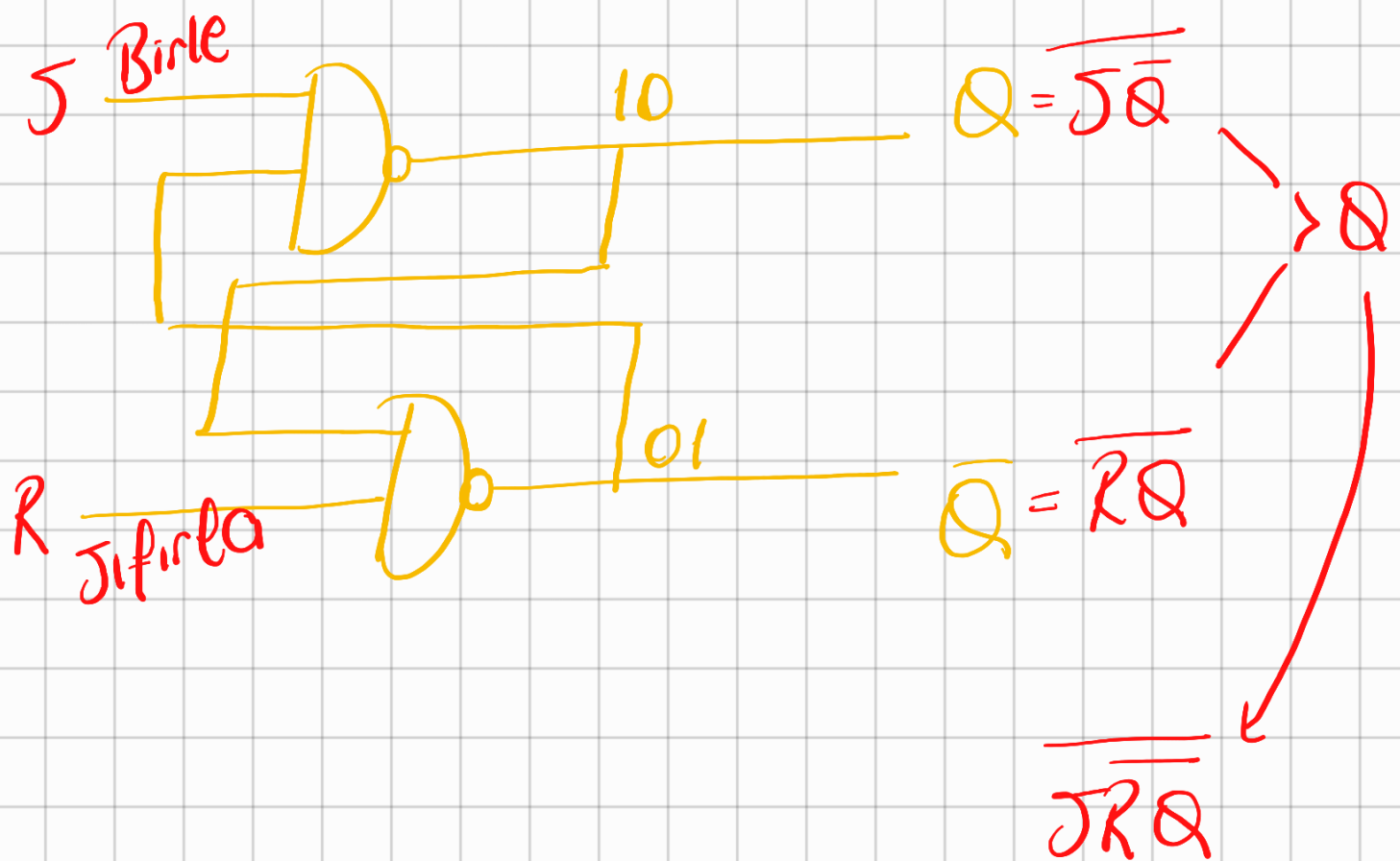




$$T = 1/F$$

$$T_{1/2} = 1/T_{1/2} = 2F$$

latch (latches)



$Q(t) = 1$ ise

$JR = 00$

$JR = 01$

$JR = 10$

$Q(t+1) = \text{tanım} \delta_{12}$

$Q(t+1) = 0$

$Q(t+1) = 1$

$$JR = 10$$

$$JR = 11$$

$$Q(t+1) = 1$$

$$Q(t+1) = Q(t) = 1$$

$Q(t) = 0$ ie

$$JR = 00 \quad Q(t+1) = \text{anmsız}$$

$$JR = 01 \quad Q(t+1) = 0$$

$$JR = 10 \quad Q(t+1) = 1$$

$$JR = 11 \quad Q(t+1) = Q(t) = 0$$

flap flopların haricinde latchesler
de kullanılır

① Ardışıl Devre Analiz yöntemleri
inceleme Süreci

① FF (Durum saklama birimleri)
in süre çıkışlarına ait

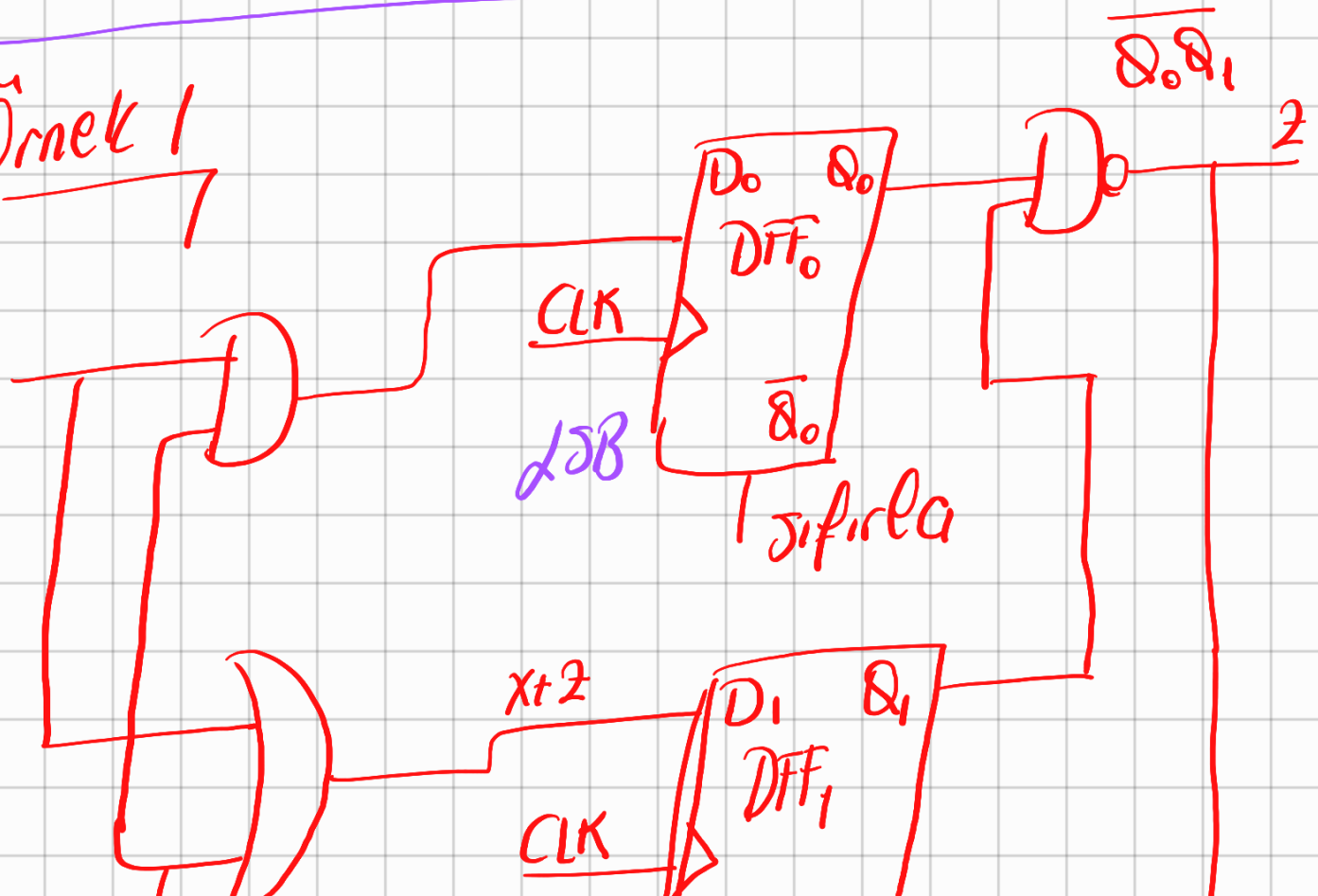
ve devre
boole fonksiyonları devreye
bakarak yazılır

② FF(dsb) peşis fonksiyonları
kullanarak bir don-
raki durum fonksiyonları
elde edilir

③ Durum tablosu oluşturulur

④ Durum diyagramı çizilir

Örnek 1





$$D_0 = X, Z = X_1(Q_1, Q_0)$$

$$D_1 = X, Z = X + (\overline{Q_1}, Q_0)$$

$$Z = (\overline{Q_1}, Q_0)$$

İmpedel durum	İindiki durumlar	
	Q_1	Q_0
d_0	0	0
d_1	0	1
d_2	1	0
d_3	1	1

$(Y) (t+1)$

FF girişleri

Bir sonraki durumlar

$X=1$

$X=0$	$X=1$	$X=0$	$X=1$
$D_1 D_0$	$D_1 D_0$	$Q_1^+ Q_0^+$	$Q_1^+ Q_0^+$
10	11	10	11
10	11	10	11
10	11	10	11
00	10	00	10

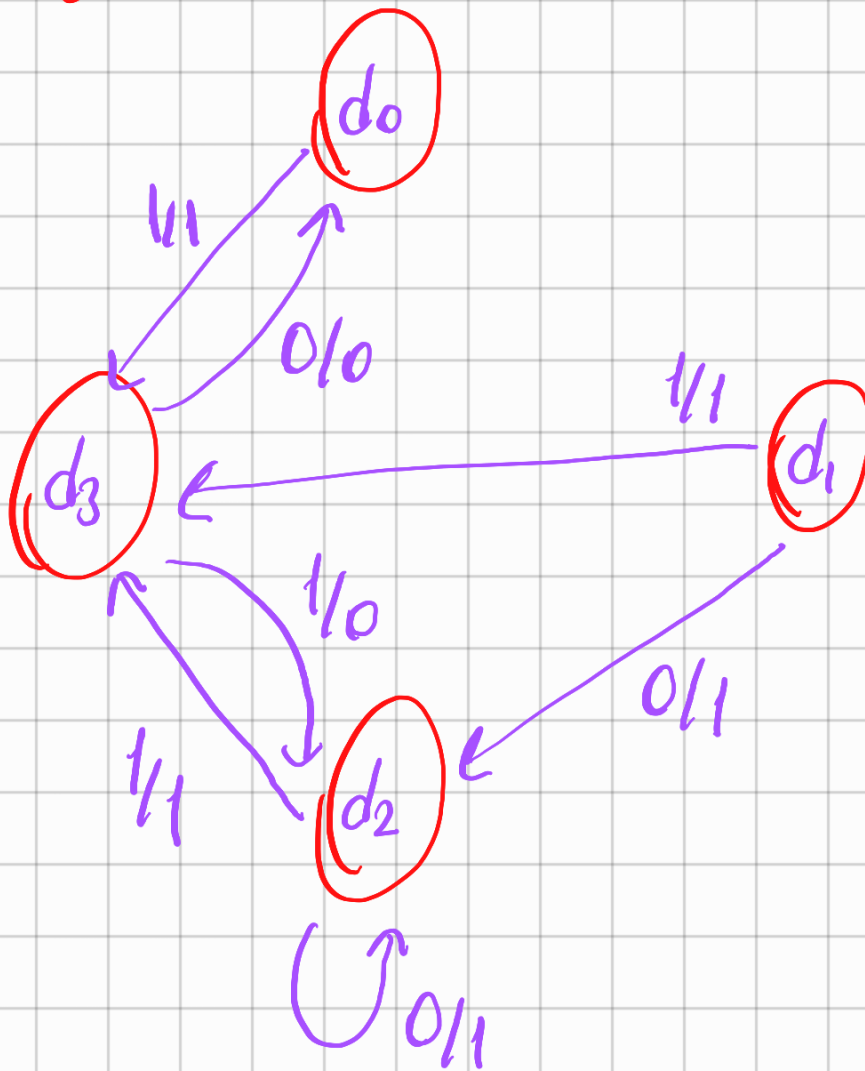
$$Y=0$$

Giriş
1
1
1
0

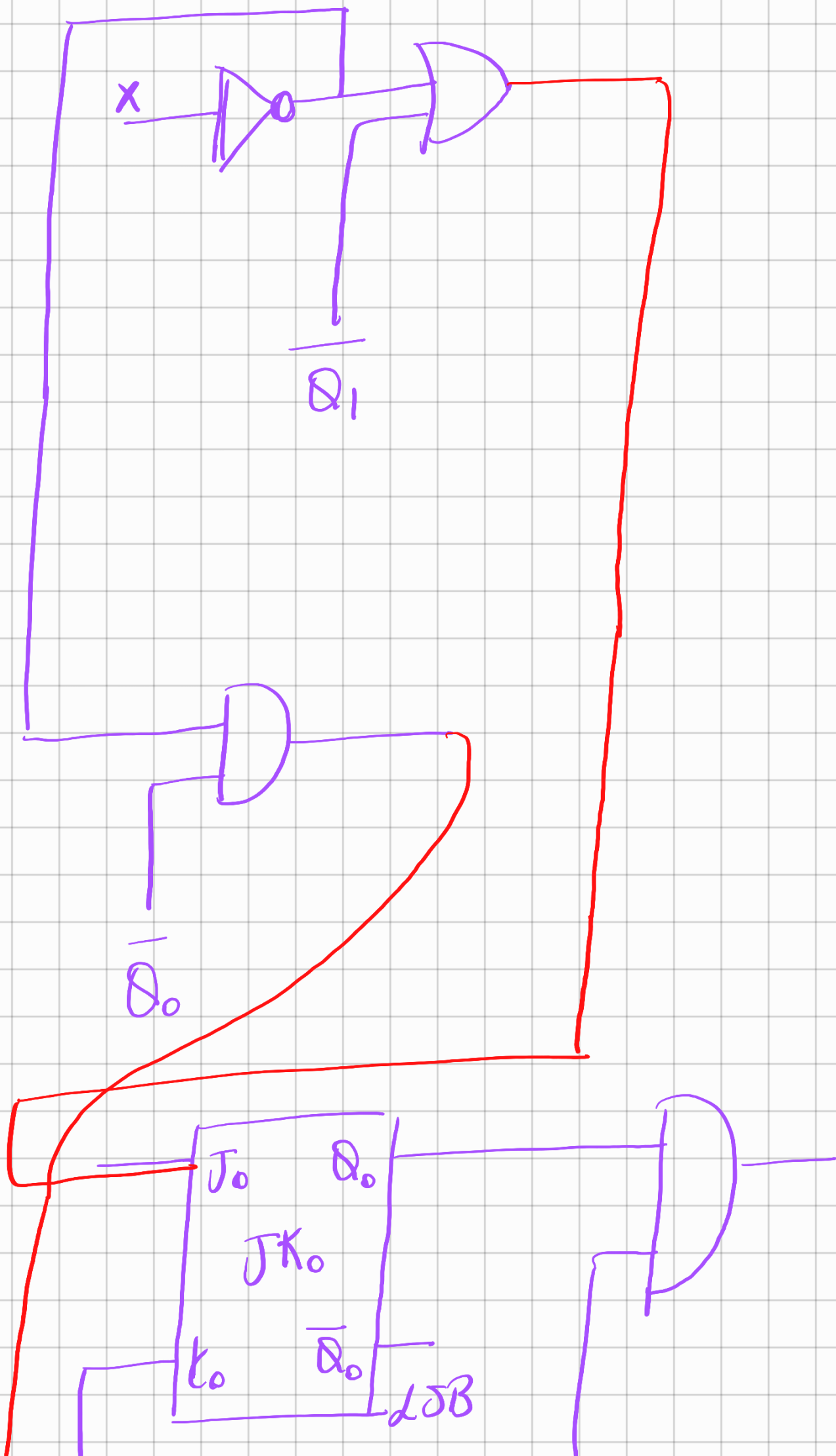
$(y) t$	$Y(t+1)$		Çıkış
Şimdiki durum	Bir sonraki durum		
	$X=0$	$X=1$	
d_0	d_2	d_3	1
d_1	d_2	d_2	1

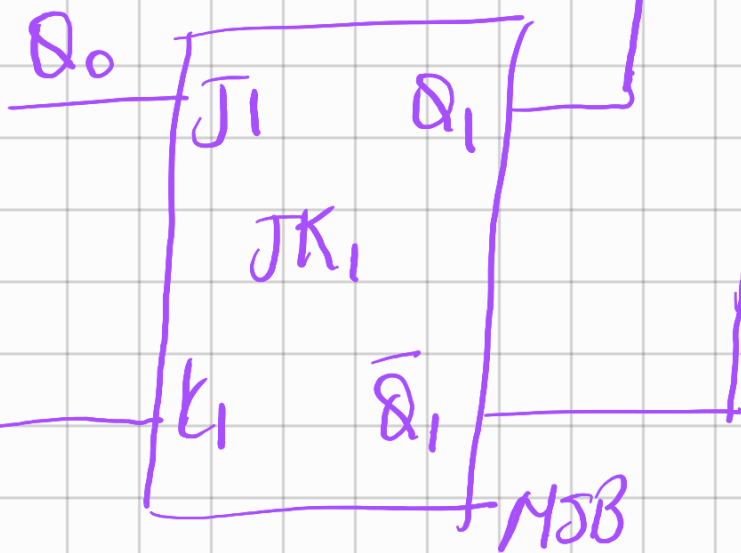
d_1	d_2	d_3	
d_2	d_2	d_3	1
d_3	d_0	d_2	0

S/G



X/G Durum Diyagramı





2 FF $\rightarrow$ 4 durum

$$J_0 = \bar{X} + \bar{Q}_1$$

$$K_0 = Q_1$$

$$J_1 = Q_0$$

$$K_1 = \bar{X} \bar{Q}_0$$

$$Z = \bar{Q}_1 Q_0$$

İmpedel
durum

İmdiki
durumlar

Q_1 Q_0

d_0	0	0
d_1	0	1
d_2	1	0
d_3	1	1

FF pirişleri

	$x=0$	$x=1$
Jiki	Joko	Jiki
01	10	00
10	10	10
01	11	00
10	11	10

$y(t+1)$
Bir sonraki durumlar

$x=0$	$x=1$
Q_1	Q_1^+
0	0
1	1

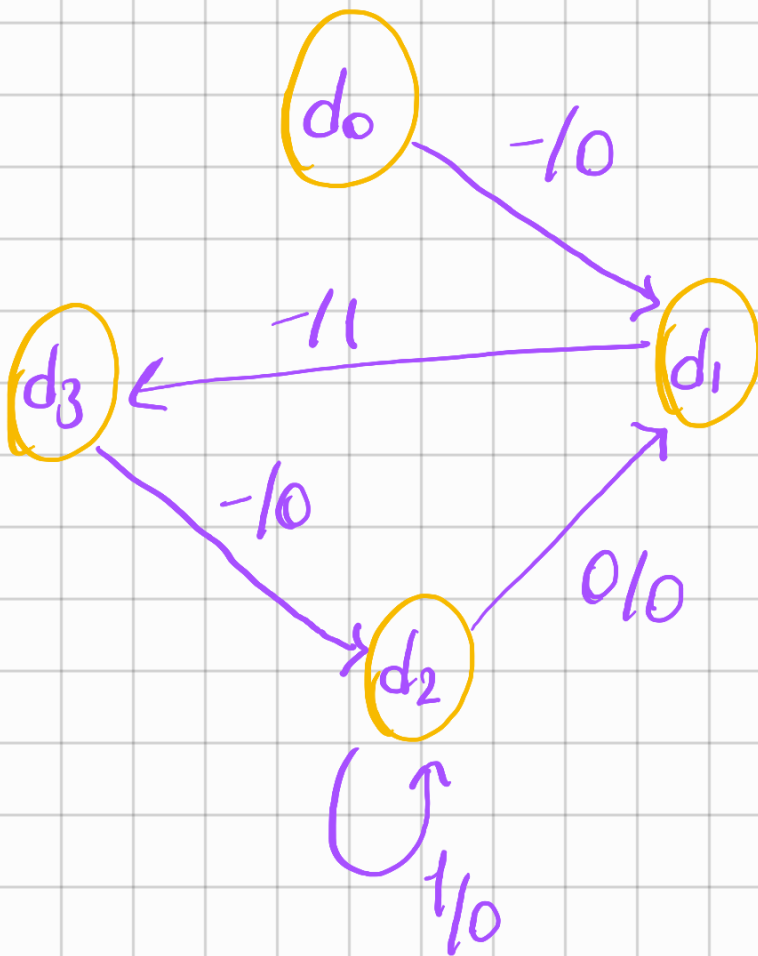
1	1	1	1
0	1	1	0
1	0	1	0

J	K	Bir sonraki durumlar
0	0	Q
0	1	0
1	0	1
1	1	$\bar{Q}$

Gelecek (2)
0
1
0
0

Q(t+1)

$(y) \& (t)$ Şimdiki durum	$y \& (t+1)$ Bir sonraki durum		Çıkış (z)
	$x=0$	$x=1$	
d_0	d_1	d_1	0
d_1	d_3	d_3	1
d_2	d_1	d_2	0
d_3	d_2	d_2	0



Durum Diyagramı

Don't

